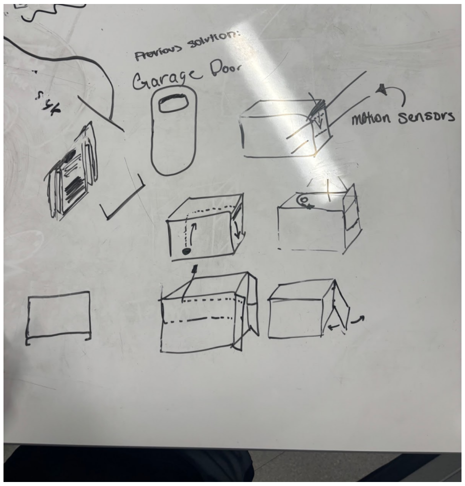
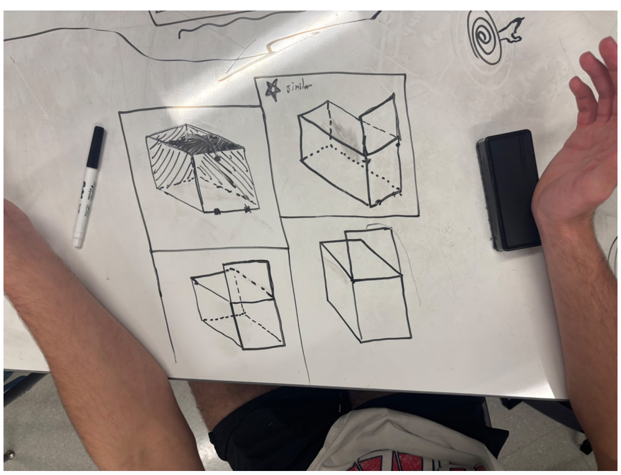
BRAINSTORMING:

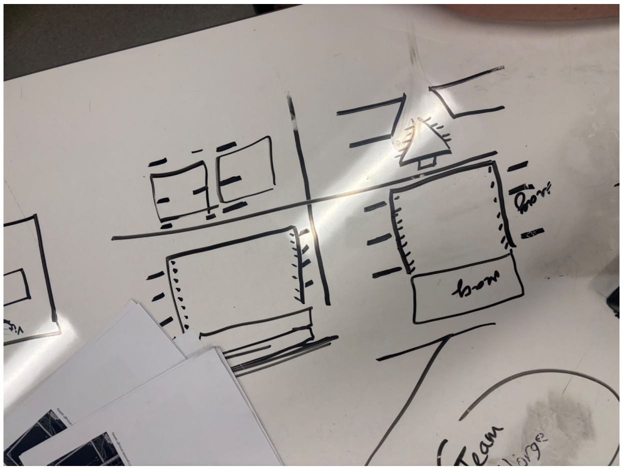
Connor's Ideas:



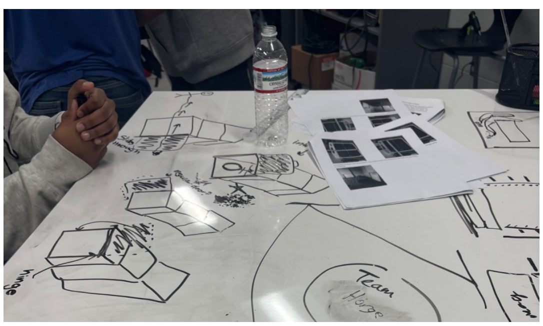
George's Ideas:



Joseph’s Ideas:



Madison’s Ideas:



First, our team brainstormed multiple ideas separately. After we had each come up with about eight ideas each, we collaborated as a team to choose four good options from the 32 ideas. We then created a design matrix where we compared four designs against criteria and constraints that were weighted by importance of constraints. In an interview with the stakeholders, we chose our design constraints that we would compare our design against, weighing each design and scoring them to figure out which design would be the most effective. We took the highest scoring number and tried to make a prototype using that option as a basis.

Joseph	Weight as decimal	Sliding Guillotine Door w/ Clamps	String Pulley Guillotine	Vertical Hinge Doors	Electromagnet
Durability & Strength	0.35	10	5	4	3
Reliability	0.25	7	5	4	3
Safety	0.15	8	10	9	10
Ease of Use	0.1	9	8	10	2
Material Suitability	0.1	9	7	5	2
Cost & Feasibility	0.05	8	7	7	1
	total score	8.65	6.35	5.6	3.75
George	Weight as decimal	Sliding Guillotine Door w/ Clamps	String Pulley Guillotine	Vertical Hinge Doors	Electromagnet
Durability & Strength	0.35	9	4	9	8
Reliability	0.25	8	8	8	8
Safety	0.15	5	5	5	5
Ease of Use	0.1	9	3	8	6
Material Suitability	0.1	6	6	5	3
Cost & Feasibility		9	7	6	1
	total score	7.85	5.4	7.5	6.5

Connor	Weight as decimal	Sliding Guillotine Door w/ Clamps	String Pulley System	Vertical Hinge Doors	Electromagnet
Durability & Strength	0.35	8	6	3	6
Reliability	0.25	5	6	4	8
Safety	0.15	6	7	2	6
Ease of Use	0.1	9	5	2	8
Material Suitability	0.1	7	7	4	5
Cost & Feasibility	0.05	7	6	3	1
	total score	6.9	6.15	3.1	6.35
Madison	Weight as decimal	Sliding Guillotine Door w/ Clamps	String Pulley Guillotine	Vertical Hinge Doors	Electromagnet
Durability & Strength	0.35	8	4	4	7
Reliability	0.25	6.5	4.5	5	6
Safety	0.15	6	8	2	8
Ease of Use	0.1	8	6	5	7.5
Material Suitability	0.1	7	7	3.5	4
Cost & Feasibility	0.05	6	5.5	3.5	1
	total score	7.125	5.3	3.975	6.35
Step 3: check out your					

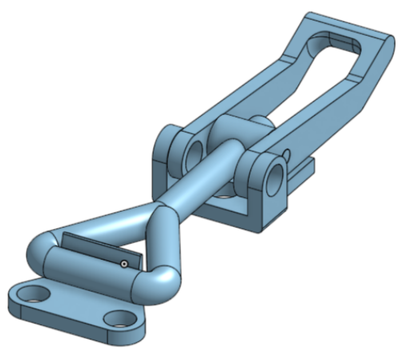
team's results!					
		Sliding Guillotine Door w/ Clamps	String Pulley Guillotine	Vertical Hinge Doors	Electromagnet
	Raw Total	30.525	23.2	20.175	22.95
	Total (out of 10)	7.63125	5.8	5.04375	5.7375

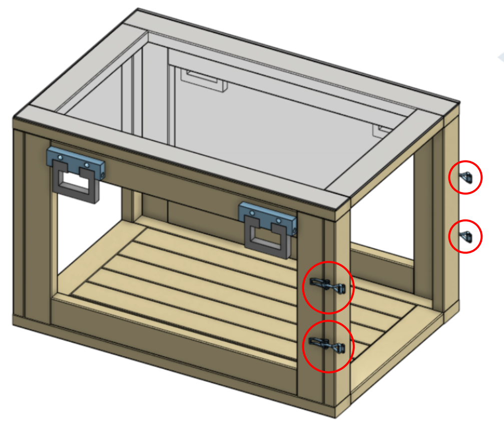
Once we created the decision matrix, we started building the prototype for the sliding guillotine door with clamps using cardboard as a substitute for the steel that we plan on using currently.

DESIGN SELECTION →

Box Connection System:

Toggle Clamp:





As of now we have decided on this toggle clamp for now; however, we are currently discussing switching to another mechanism that synthesizes better with the box frame. This is the best option as of now and we currently have a prototype for this on the current prototype. We attach one side of the toggle clamp to one of the boxes and the other to the other box. Although this doesn’t make the boxes interchangeable, the boxes will be heavily secured under the tension created by locking the clamps in place. The biggest issue that we are having with this is that the current toggle clamps can endure little tension. On top of that, in order to ensure that there is enough stability across the entire box, we have to attach 4 clamps to each spot so that there’s enough tension to hold the boxes together without the weight of the Komodos overcoming the clamps. With this in mind, we started brainstorming other ideas that would better suit the box as well as fix the current issue that we are having with this design.

Butterfly Twist Clamp:

This is the secondary prototype in case we need to pivot from either of the previous designs. The reason that we decided on these as a secondary option is because of the strength of these clamps along with the compatibility of these with various other designs. These kinds of clamps have been proven to be extremely durable and effective in other situations.

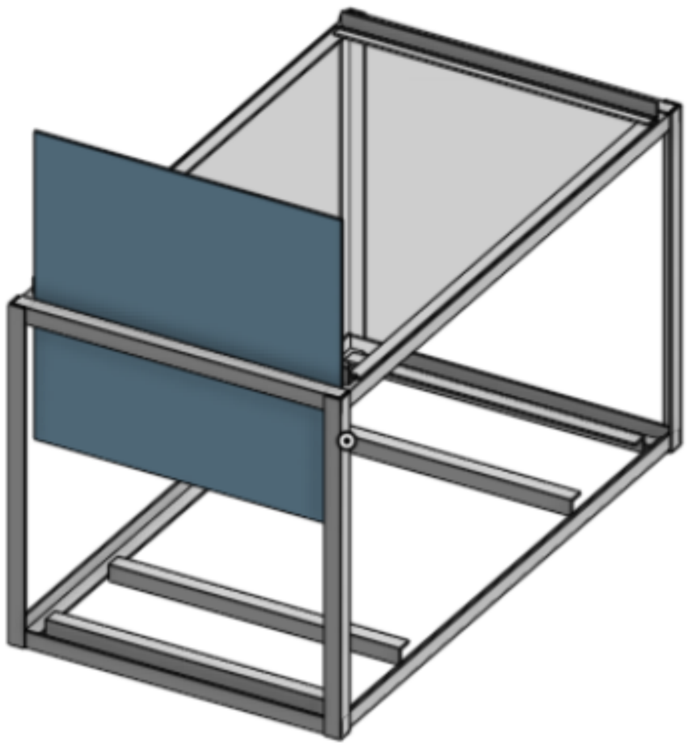
Other real world uses:

Source: https://www.robosource.net/robot-cases

These kinds of boxes are very prevalent in the world of robotics which is something that our school offers. In this system, the clamp is attached to the bottom of the box and as the lid is placed on top, the hinges on the clamp allow the clamp to grab hold of the top of the box and clamp down. Another thing is that these clamps have a proven stability because each year they must travel to Dallas, TX on an airplane and must contain a robot inside. If it comes to anything, we will most likely decide on these clamps because of their proven durability and versatility. These clamps are also proven to hold more weight than the toggle clamps and we wouldn’t be forced to use more than one clamp on each side of the boxes which would be optimal.

The next part of the project that we are designing is the guillotine door.

Guillotine Door:

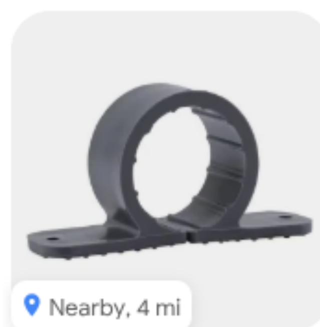


The final aspect of our design is the locking mechanism for the guillotine door. Our brainstorming yielded several variations, primarily differing in how the door secures into place—either by sliding into a slot in the floor or through an external locking mechanism. Testing will determine which method offers the best combination of security and ease of use.

Through a structured and iterative process, we successfully evaluated multiple design options, ultimately selecting a robust and functional concept. The Sliding Guillotine Door with Clamps was chosen due to its superior strength, reliability, and usability, ensuring a well-justified and effective final design, but also its adaptability and flexibility in terms of ease of modifications.

New Design Selection:

After a meeting with our teacher, we were made aware of the fact that our budget had been cut from $300 to about $50-$100. This was especially problematic because the other group working on the design of the boxes originally budgeted for their design to be around $600-$800. Knowing this, we had to work diligently to reduce our budget as much as possible. Talking with Bill Rodriguez we were able to come up with a solution for how we can cost effectively design a solution for attaching the three boxes to each other. Below are two of the different options that we could choose from:

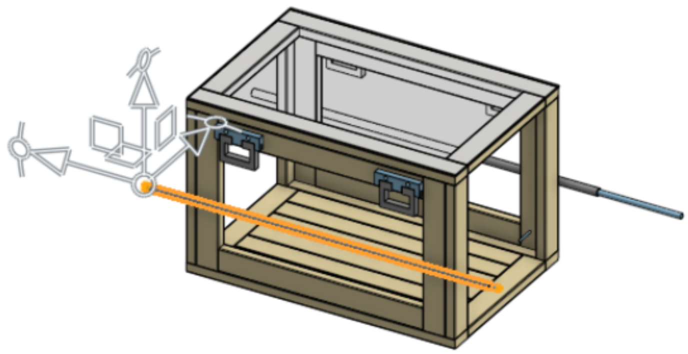


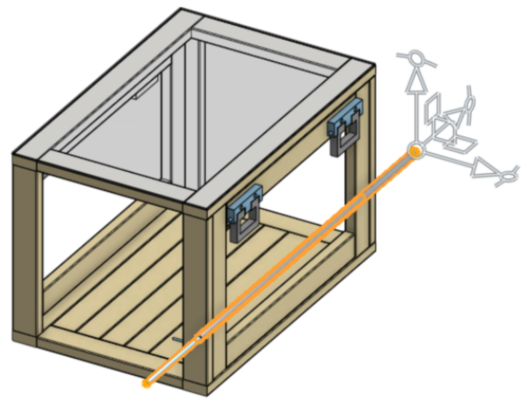
Sources:

Left: Carlon PVC Conduit Clamp 3/4-in. 5-Pack

Right: HoldRite 5 -Pack 1-in to 1-in dia Plastic Standard pipe clamp

We chose this because it is relatively cheap, costing around $20 for a 10-pack of clamps. This is incredibly effective in terms of keeping the budget more reasonable. In terms of total pricing, this is the only thing that we are buying. The door is going to be handmade by a student in our class and we plan on hand cutting the holes and slits for the handle and door locking system.



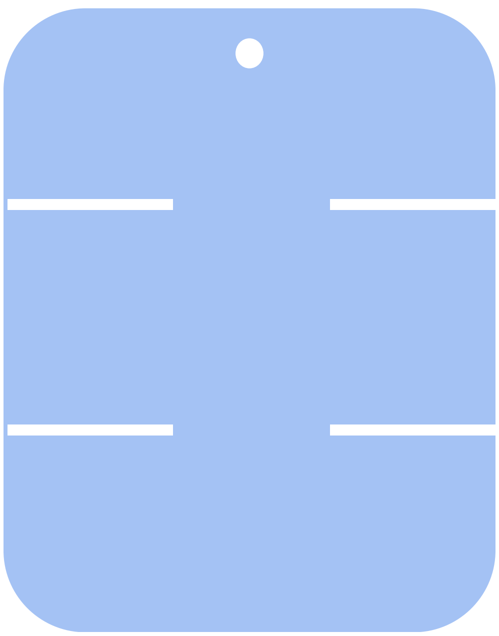


The next thing that we need to focus on in terms of the design is the guillotine door. Every little detail needs to be correct so that the door works properly and effectively enough.

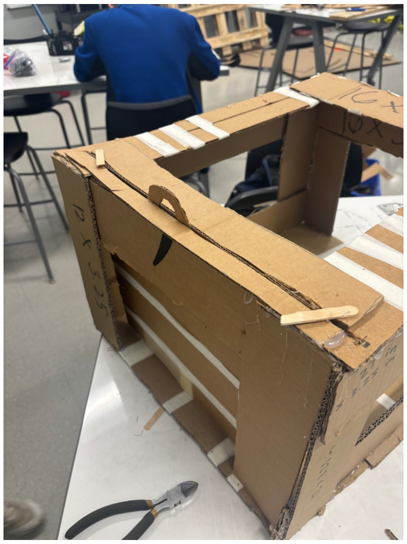
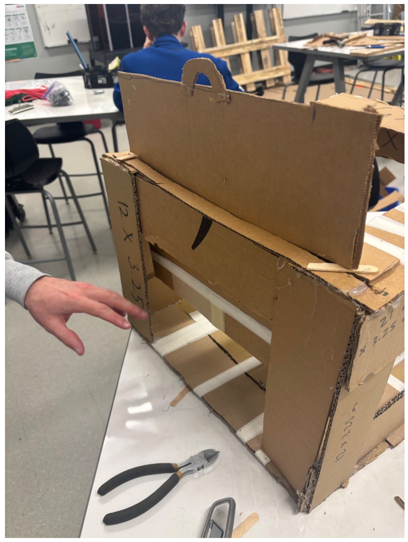
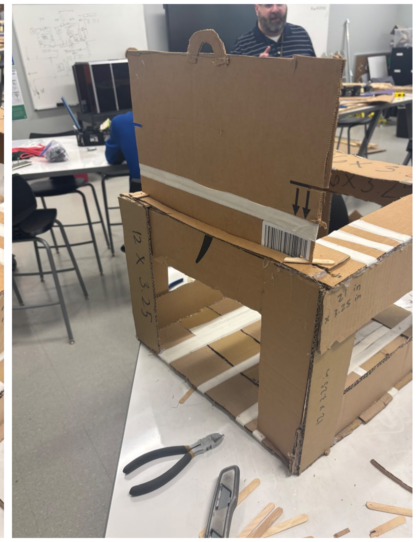
Guillotine Door:

Source: https://securitybosspetdoors.com/classic-guillotine-kennel-door/

Above is the design that we plan on using for our guillotine door. Although it will look different from this design, it is a similar design to the door that we plan on using. The only difference may be the shape of the door which might be in the shape of a rectangle with curved corners. As shown below, this is the idea for what we expect our door to look like.

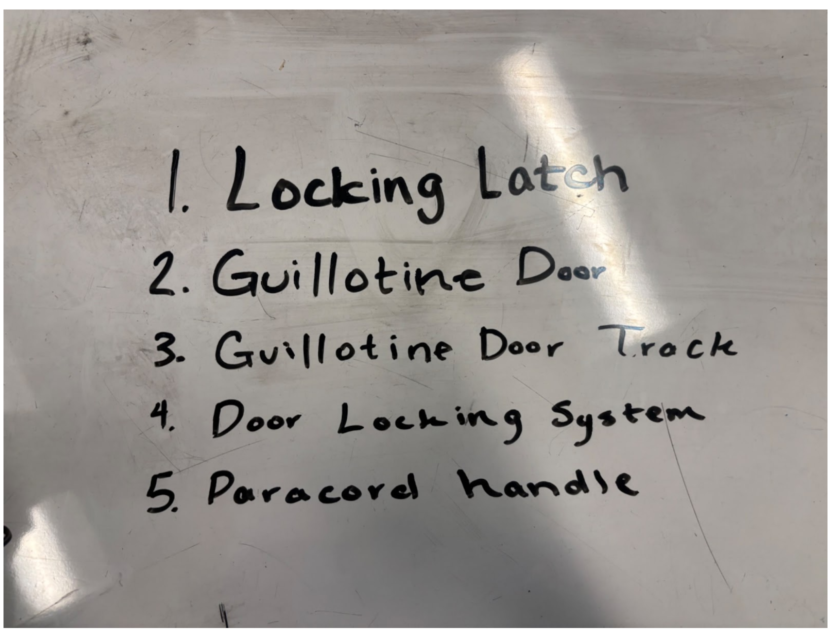


Above is how our guillotine door is expected to look. This prototype worked extremely well when prototyped with a cardboard box and popsicle sticks. Below are three of the pictures of the prototype:

The prototype can be seen above working on three different settings using the cardboard prototype. The idea is that this design will be able to hold the door at three different heights for transportation of the Komodo dragon to move between boxes.

In terms of higher-priority focuses, we made a list of the things that we need to complete:



This list is not set in stone, but with about ~25 days until the due date for this project we need to start to focus on the more important aspects of our project so that we have something to present.